Sparse autoencoders reveal organized biological knowledge but minimal regulatory logic in single-cell foundation models

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Abstract

Whether single-cell foundation models encode causal regulatory logic, rather than statistical co-expression, remains unclear. We trained TopK sparse autoencoders on residual-stream activations from every layer of Geneformer V2-316M and scGPT whole-human, producing atlases of 82,525 and 24,527 features, released as interactive platforms. The features are biologically organized: 29–59% annotate to standard ontologies and form co-activation modules. Under a capacity-matched comparison they are not hidden from linear decomposition—truncated SVD needs rank 250–403 to match their reconstruction quality—but give a sparse parameterization supporting per-feature intervention. Ablating one feature disrupts the genes it fires on by 35–67×, but not the wider process it is annotated with. Against CRISPRi perturbation in four cell lines, only 1 of 66 transcription factors shows target-specific responses. Benchmarking against differential expression on the same cells, established network inference and random controls explains why: that assay measures 6,546 genes and the median perturbed factor has one curated target among them, so target-specific recovery is undefined for most of the panel. On perturbation data covering the transcriptome the features do enrich for curated regulons, for 3 of 13 factors against 1 of 9 perturbations of non-regulatory genes—the direction specificity predicts, on a panel too small to establish it ($p = 0.62$). Whether these representations carry a small amount of factor-specific regulatory information is therefore not settled by currently available assays, and how much of a regulon a screen can measure is essential to interpreting any claim about what a model encodes.

1 Introduction

Single-cell foundation models (scFMs) such as Geneformer [1] and scGPT [2] are used for cell-type annotation, perturbation-response prediction, and gene-network inference. They are trained on millions of transcriptomic profiles and learn contextual gene representations without explicit supervision on regulatory relationships. A central question is whether these representations encode *causal regulatory logic*—the directed relationships between transcription factors (TFs) and their target genes—or only the *statistical co-expression* that correlates with, but does not constitute, regulation.

At the level of attention weights this question has been addressed by a companion study [3], which found that attention captures co-expression rather than regulatory signal: trivial gene-level baselines outperform attention edges, pairwise scores add zero incremental predictive value, and causal ablation of putatively regulatory heads produces no behavioural effect. Attention weights, however, are one view among several. The residual stream—the running sum of all layer outputs—may encode richer structure than attention patterns alone reveal.

The superposition hypothesis [4] offers a framework for this richer structure. When a model must represent more concepts than it has dimensions, it encodes features as nearly-orthogonal directions in activation space and activates only a sparse subset of them per input. Sparse autoencoders (SAEs) [5, 6, 7, 8] resolve superposition by learning an overcomplete dictionary with a sparsity constraint, decomposing dense activations into interpretable features. In large language models, SAE pipelines have scaled to billions of features [9, 10]; they have not been systematically applied to biological foundation models, where the "concepts" are genes, pathways, and regulatory programs rather than linguistic entities.

Here we present a systematic SAE-based interpretability analysis of single-cell foundation models. We train TopK SAEs [11] on per-position residual-stream activations from all 18 layers of Geneformer V2-316M and all 12 layers of scGPT whole-human [2], producing atlases of 82,525 and 24,527 features respectively. The two models differ substantially in architecture (Geneformer's rank-value tokenization vs. scGPT's dual gene-ID plus binned-value embeddings, 18 vs. 12 layers, d=1,152 vs. d=512, bidirectional masked-token vs. specialized causal-attention generative training) and training data (30M vs. 33M cells), yet we apply an identical analytical pipeline to both.

Two questions organize the analysis. The first is descriptive: what biological structure do these dictionaries contain, how is it organized across layers, and how much of it is accessible to standard linear decomposition of the same activations? The second is causal: do the features encode which genes a transcription factor actually regulates? Answering the second question quantitatively requires a reference point, because a low recovery rate is uninformative unless one knows how much of the regulatory relationship is recoverable from the same measurements by any method. We therefore benchmark the SAE against a perturbation-aware empirical ceiling— differential expression of the CRISPRi knockdown itself—and against established regulatory-inference baselines evaluated in an identical framework, and we replicate the measurement in four cell lines.

To make these results accessible to the community, we release both atlases as interactive web platforms—the Geneformer Feature Atlas (`https://biodyn-ai.github.io/geneformer-atlas/`) and scGPT Feature Atlas (`https://biodyn-ai.github.io/scgpt-atlas/`)—enabling exploration of over 107,000 features across 30 layers of two leading single-cell foundation models.

2 Results

An SAE feature atlas of Geneformer

We extracted per-position residual stream activations from all 18 layers of Geneformer V2-316M, processing 2,000 non-targeting control cells from the Replogle genome-scale CRISPRi dataset [12] and obtaining 4,056,351 token positions per layer (mean 2,028 genes per cell; 336.4 GB in total). The control condition in this resource is defined by perturbation label rather than by cell line, so the cohort spans all four lines it contains—591 Jurkat, 581 K562, 567 RPE1 and 261 HepG2 cells (Supplementary Table S1)—and the dictionaries below are learned from that multi-line control population. For each layer we trained a TopK sparse autoencoder with a 4× overcomplete dictionary (4,608 features from 1,152 input dimensions) and k=32 sparsity, subsampling 1 million positions per layer for training efficiency (Methods).

Table 1 presents training and annotation statistics for all 18 layers (Fig. 1). Variance explained peaks at layers 3–4 (85.2–85.3%) and generally declines thereafter to 76.8% at layer 17, with a brief recovery at layers 10–11, indicating that later-layer representations become progressively more distributed and harder to compress with a fixed-size dictionary. Dead features—those never activating

Table 1: **Geneformer SAE training and annotation statistics.** Each SAE uses a 4× overcomplete dictionary (4,608 features) with TopK k=32 sparsity. VarExpl = variance explained. Dead = 4,608 − Alive. Aligned = features with $|\cos| > 0.5$ against any of the leading 50 principal axes; see Supplementary Table S14 for the capacity-matched comparison. MeanCos = mean absolute pairwise cosine between decoder vectors. Ann% = fraction of alive features with ≥1 significant enrichment (FDR < 0.05). Enrichments = unique (feature, term) pairs. Per-database breakdown in Supplementary Table S2.

Layer	VarExpl	Alive	Dead	Aligned	Non-aligned	MeanCos	Ann%	Enrichments
0	83.9%	4,608	0	41	4,567	0.033	58.6%	23,959
1	84.6%	4,606	2	27	4,579	0.033	57.4%	23,131
2	84.8%	4,601	7	29	4,572	0.034	55.5%	23,383
3	85.2%	4,595	13	12	4,583	0.035	56.4%	21,922
4	85.3%	4,583	25	13	4,570	0.037	53.9%	19,910
5	84.6%	4,576	32	8	4,568	0.036	52.1%	17,865
6	83.3%	4,580	28	5	4,575	0.035	49.1%	16,716
7	81.4%	4,591	17	6	4,585	0.036	47.7%	15,470
8	80.4%	4,586	22	3	4,583	0.038	45.4%	15,679
9	80.2%	4,595	13	4	4,591	0.036	50.3%	16,925
10	81.0%	4,602	6	7	4,595	0.035	55.8%	19,587
11	81.6%	4,598	10	4	4,594	0.037	56.2%	19,943
12	81.0%	4,592	16	3	4,589	0.037	53.2%	19,105
13	80.1%	4,583	25	3	4,580	0.038	50.7%	17,338
14	80.0%	4,568	40	2	4,566	0.040	50.6%	17,719
15	78.8%	4,543	65	5	4,538	0.038	49.0%	15,571
16	76.9%	4,538	70	5	4,533	0.037	54.7%	16,080
17	76.8%	4,580	28	12	4,568	0.039	47.0%	15,764
Total		**82,525**	**419**	**189**	**82,336**			**336,067**

on 100K held-out positions—increase from 0 at layer 0 to 70 at layer 16, with a trough at layers 9–11 (6–13 dead) suggesting a mid-stack revival of structured representations. Dictionary directions are well separated throughout: the mean absolute cosine similarity between decoder columns ranges from 0.033 to 0.040, close to the theoretical floor for this many unit vectors in 1,152 dimensions (Methods).

The atlas comprises **82,525 alive features** across all 18 layers, of which **43,241** (52.4%) receive at least one significant ontology annotation against five databases (Gene Ontology Biological Process [13], KEGG [14], Reactome [15], STRING [16], and TRRUST [17]).

An SAE feature atlas of scGPT

To test whether the organization observed in Geneformer generalizes across architectures, we applied an identical SAE pipeline to scGPT whole-human [2]—a model trained on 33 million cells with fundamentally different design choices: a dual input of gene-identity tokens plus binned expression-value embeddings (51 bins by default, in contrast to Geneformer's rank-value tokenization), 12 transformer layers rather than 18, d_{model}=512 rather than 1,152, and a generative causal-attention training objective rather than bidirectional masked-gene modelling.

Table 2: **scGPT SAE training and annotation statistics.** 4× dictionary (2,048 features), TopK k=32, 3.56M positions/layer. Notation as in Table 1.

Layer	VarExpl	Alive	Dead	MeanCos	Ann%	Modules
0	92.0%	2,027	21	0.038	32.7%	6
1	93.0%	2,035	13	0.041	30.8%	6
2	93.2%	2,038	10	0.040	28.7%	5
3	93.2%	2,045	3	0.041	31.6%	7
4	93.5%	2,048	0	0.041	30.4%	6
5	92.1%	2,048	0	0.042	29.6%	7
6	90.3%	2,048	0	0.046	28.9%	7
7	85.7%	2,048	0	0.049	32.4%	6
8	86.3%	2,048	0	0.046	28.9%	7
9	86.9%	2,047	1	0.045	33.9%	5
10	87.4%	2,047	1	0.044	31.7%	7
11	88.6%	2,048	0	0.043	32.0%	7
Total		**24,527**	**49**			**76**

We extracted per-position residual-stream activations from all 12 layers, processing 3,000 diverse Tabula Sapiens cells (1,000 immune across 43 cell types, 1,000 kidney, 1,000 lung) and obtaining 3,561,832 token positions per layer. For each layer we trained a TopK SAE with a 4× overcomplete dictionary (2,048 features from 512 input dimensions) and k=32 sparsity.

Table 2 presents training statistics for all 12 scGPT layers. Reconstruction quality is higher than for Geneformer: variance explained ranges from 85.7% (L7) to 93.5% (L4), mean 90.2% against Geneformer's 81.7%. Dead features are near zero: 49 across all 12 layers (0.2%), compared with 419 in Geneformer (0.5%). The atlas comprises **24,527 alive features**, of which **7,595** (31.0%) are annotated against the same five databases. These two atlases are built from different input distributions, so the comparison between them is addressed separately under matched inputs below.

Biological annotation follows a U-shaped layer profile

Annotation of each feature's top-20 genes against five databases (Fisher's exact test, Benjamini–Hochberg FDR < 0.05) reveals a U-shaped profile across layers (Fig. 2). Annotation rates are highest at layers 0–1 (57–59%), decline to a minimum at layer 8 (45.4%), recover to 55–56% at layers 10–11, and decrease again at layers 16–17 (47–55%). The scGPT atlas shows a weaker but qualitatively similar pattern across its 12 layers, from 28.7% (L2) to 33.9% (L9).

For Geneformer this pattern admits a functional reading in four zones.

Early layers (0–4): molecular machinery. Features map cleanly onto existing ontology terms, with the highest GO BP (8,500–10,150 enrichments per layer), KEGG (2,050–2,650), and Reactome (9,200–11,000) counts. STRING protein–protein interaction associations peak here (248–302 per layer), as do TRRUST TF associations (133–164). Representative features encode textbook biological programs (Table 3).

Middle layers (5–9): abstract computation. Annotation rates drop below 50% at layers 6–8, consistent with intermediate representations that are harder to map onto a single ontology term.

Table 3: **Representative SAE features.** Top genes = highest mean activation magnitude. Databases = ontology sources with significant enrichments.

Feature	Top Genes	Biological Identity	Databases
Layer 0 — Molecular Programs			
F3717	CDK1, CDC20, DLGAP5, PBK	Cell cycle (G2/M)	GO, KEGG, React., STRING
F3607	RRM1, E2F1, MCM4, MCM6, RAD51	DNA replication / repair	GO, KEGG, React., STRING, TRRUST
F1116	ARL6IP4, SQSTM1, JAK1	B cell activation	GO, KEGG, Reactome
F4536	DYNC1H1, TLN1, MYH9, SPTAN1	Cytoskeleton / focal adhesion	GO, KEGG, Reactome
F2829	TGFB1, PKN1, GADD45A, ZBTB7A	MAPK/TGFβ signaling	GO, KEGG, React., STRING, TRRUST
F1573	MKI67, CENPF, TOP2A, CDK1, AURKB	Mitosis / chr. segregation	GO, KEGG, React., STRING
Layer 11 — Integrative Programs			
F2035	*(cell diff. genes)*	Cell diff. (neg. reg.)	GO, Reactome
F3692	*(ERAD pathway genes)*	ER-associated degradation	GO, KEGG, Reactome
F3933	*(signaling genes)*	Intracell. signaling (neg. reg.)	GO, Reactome
F2936	*(mitochondrial genes)*	Mitochondrion organization	GO, KEGG, Reactome

GO BP falls to 6,600–7,700 and STRING associations to 181–216 per layer.

Mid-late layers (10–12): re-specialization. Annotation rates recover to 53–56%, with GO BP returning to 8,200–8,800 and KEGG to 1,750–2,100. Features shift from molecular processes to integrative cellular programs such as cell differentiation, intracellular signaling, and organelle organization.

Terminal layers (15–17): prediction-focused. A second annotation decline (47–55%), the largest number of dead features (28–70), and the most distributed representations. Features at these layers respond broadly to perturbations but with less specificity than mid-layer features.

Features are rebuilt at every layer rather than carried forward

To understand how features relate across layers, we tracked decoder weight vector similarity from layer 0 features to all subsequent layers. Features are overwhelmingly layer-specific: only 2–3% of features at any layer match a feature at the adjacent layer. From layer 0 the decay is steep: 114 matches at L1 (2.5%), 67 at L4 (1.5%), 10 at L8 (0.2%), and effectively zero beyond L10 (Supplementary Table S3). No layer 0 feature survives past layer 11.

Feature persistence anti-correlates with biological content. Of layer 0 features, 98.2% are transient (present in at most three layers), with 59.6% annotated and a mean of 5.4 enrichment terms. The rare moderate-persistence features (1.8%, present in 4–10 layers) have only a 3.7% annotation rate and 0.1 mean enrichments. Biological content is carried by features rebuilt at every layer, not by a persistent scaffold.

Features organize into co-activation modules

To map the relational structure among features we constructed co-activation graphs at each of the 18 layers using pointwise mutual information (PMI [18]) over the 4 million token positions, and identified communities with the Leiden algorithm [19] (Methods).

Across all layers we identified **141 modules** (6–12 per layer) covering **96.0–99.5%** of alive features (Supplementary Table S4). This is not a trivial outcome: with k=32 sparsity out of 4,608 features, random co-activation would produce a connected but undifferentiated graph; instead the partition yields well-separated communities with distinct biological identity.

Module count peaks at layer 5 (12 modules), suggesting maximal functional compartmentalization at early-to-mid processing, and PMI edge density declines with depth (446K at L0 to 328K at L16), paralleling the declining variance explained (Fig. 3). At layer 0 six modules correspond to cell cycle and DNA replication, immune signaling, metabolism, translation, protein quality control, and cytoskeleton and adhesion. At layer 11 eight modules shift toward integrative cellular programs: cell differentiation, intracellular signaling, mitochondrial organization, vesicular transport, chromatin and transcription, protein modification, cell motility, and stress response.

The scGPT atlas exhibits the same modular organization: 76 modules across 12 layers (5–7 per layer), with a mean coverage of 96.3%. Despite the smaller dictionary (2,048 against 4,608), module count per layer is comparable (6.3 against 7.8 mean), suggesting that the number of biological programs an SAE resolves scales sub-linearly with dictionary size.

Because a single Leiden resolution is a choice rather than a ground truth, we swept $\gamma \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$ at layers 0 and 11 (Supplementary Table S11); the graph dimensions and memory footprint of the co-activation construction are in Supplementary Table S10. The partition is not stable in a strong partition-agreement sense—at γ=0.5 the graph collapses into two large communities and at γ=2.0 it fragments into 48–56, with adjusted Rand indices of 0.17–0.26 between neighbouring resolutions. What is stable is module *identity*: the same biological themes are recognizable throughout the range, with larger γ splitting one program into finer subprograms. Module counts should therefore be read as the resolution-1.0 cut of a hierarchical community structure.

Cross-layer information highways

Given that features are almost entirely layer-specific, how does biological information flow through the network? We computed cross-layer PMI between SAE feature activations, encoding the same 500,000 positions through both source- and target-layer SAEs. An information highway is a source-layer feature with at least one dependency on a target-layer feature above a PMI threshold τ.

Under a TopK constraint two features can co-activate simply because of marginal activation rates, so any fixed threshold on raw PMI risks inflating the highway count. We therefore computed a permutation null for three long-range pairs (L0→L5, L5→L11, L11→L17) by independently column-shuffling the target-layer activation matrix 20 times, recomputing the statistic, and taking the maximum across rounds of the 99th percentile of finite PMI as a per-pair null threshold. The reported count uses $\tau = \max(3.0, \tau_{\text{null}}^{p99})$; the null threshold came out at 2.46 or below for every pair, so the effective threshold is 3.0 and the highway fractions are not an artifact of the sparsity constraint.

For the long-range pairs (Supplementary Table S5, Supplementary Table S6), L0→L5 is at 98.96%, L5→L11 at 96.66%, and L11→L17 at 99.02%, with mean maximum PMI between 6.61

and 6.79 and individual connections exceeding PMI 10. Across all 17 consecutive pairs $L_i \to L_{i+1}$ the curve is near-flat in the 94.7–99.1% range (Supplementary Table S7) with no sharp drop at any transition (Fig. 4); the slight taper at L16→L17 (94.70%) is consistent with the terminal-layer specialization described above. Information flow is therefore continuous across the Geneformer stack rather than bottlenecked at particular layers—in line with layer-wise benchmarking of scFM representations in downstream tasks [20].

Biologically meaningful cascades appear among the strongest cross-layer connections (Supplementary Table S9). The mTORC1-regulation to autophagy connection (L0→L5, PMI = 9.55) recapitulates a well-known signaling axis, and protein-modification features at L11 connect to angiogenesis regulation at L17 (PMI = 10.62).

scGPT shows a different pattern (Supplementary Table S8, Supplementary Fig. S1): upstream connectivity is consistently high (86.6–95.5%), but downstream connectivity falls from 95.7% (L0→L4) to 62.9% (L8→L11), indicating a progressive concentration of information toward later layers that Geneformer does not show (97.4–99.8% downstream throughout).

Feature space geometry

Projecting decoder weight vectors with UMAP under cosine distance produces structureless point clouds at every layer. This is the expected geometry rather than a failure of the projection: decoder directions are quasi-orthogonal by construction, with a mean *signed* pairwise cosine of 0.0007 and a mean *absolute* pairwise cosine of 0.033–0.040 (Table 1), and a within-module versus between-module effect size of Cohen's $d = 0.075$. Thousands of features pack into 1,152 dimensions by spreading nearly uniformly over the hypersphere, so no distance-preserving two-dimensional embedding of the directions themselves can be informative.

We therefore visualized the co-activation network directly using a force-directed layout (Supplementary Fig. S2). Each module forms a spatially coherent community, with annotated features distributed across all communities while sparsely annotated features concentrate centrally, and the layer-11 graph resolves this structure in detail (Supplementary Fig. S3). Annotation-based projections (TF-IDF weighted ontology vectors) provide independent confirmation that the module structure reflects biological similarity (Supplementary Fig. S4).

Cell type enrichment

To connect features to cellular identity we computed, for each feature, its mean activation in cells of each type among the 3,000 Tabula Sapiens cells spanning 56 cell types across immune, kidney and lung tissue, and tested for enrichment (Fisher's exact test, Benjamini–Hochberg correction).

In the scGPT atlas, 2,028 of 2,048 features (99.0%) at layer 7 are enriched for at least one cell type, with similar coverage at all layers. The patterns are tissue-coherent: immune features activate preferentially in T cells, B cells, macrophages and dendritic cells; kidney features in proximal tubule cells and podocytes; lung features in alveolar epithelial and pulmonary endothelial cells. For the Geneformer atlas we passed the same Tabula Sapiens cells through the model and encoded them with the multi-line control dictionaries; despite the mismatch in training distribution—immortalized lines against primary tissue—the dictionaries generalize to these tissue contexts. Features of both models tile cell-identity space comprehensively, consistent with the models' training on diverse cell populations.

Batch and assay confounds

Cell-type enrichment alone does not distinguish biology from donor or sequencing-assay batch effects, a particular concern for Tabula Sapiens, which spans 24 donors and 4 sequencing chemistries. For each scGPT feature at layers $\{0, 4, 7, 11\}$ we computed the mutual information between its activation and two categorical metadata fields: tissue (3 values) and cell type (56 values) (Supplementary Table S26), tissue serving as a coarse proxy for sample-level batch because the three tissues came from different preparations. Across all four layers, mean mutual information with cell type is about 2.5× that with tissue—at layer 11, 0.0040 against 0.0016—and essentially no feature is tissue-dominant (0/1990 at L0, 0/2038 at L4, 0/1999 at L7, 1/1766 at L11). Coarse tissue batch is therefore not the primary driver of scGPT feature activation. Donor and assay identity are not recorded in the cached extraction metadata and a donor-level decomposition is beyond what these data support.

Interactive feature atlases

To enable community access to the complete atlases we developed and deployed two interactive web platforms: the Geneformer Feature Atlas (`https://biodyn-ai.github.io/geneformer-atlas/`; 82,525 features across 18 layers) and the scGPT Feature Atlas (`https://biodyn-ai.github.io/scgpt-atlas/`; 24,527 features across 12 layers). Both provide six integrated views—Layer Explorer, Feature Detail Pages, Module Explorer, Cross-Layer Flow, Gene Search, and Ontology Search—are built with React, Vite and Plotly.js, served via GitHub Pages, and require no installation.

What the dictionary adds over the principal subspace

A dictionary of 4,608 atoms and a basis of 50 principal axes are not comparable objects, so the observation that few features align with the leading axes cannot by itself establish that the information is inaccessible to linear decomposition. We therefore computed the exact eigendecomposition of the activation covariance at every layer and compared the two representations under matched conditions (Fig. 5, Supplementary Table S13 and Supplementary Table S14).

Reconstruction-matched capacity. Truncated SVD requires rank **250–403** to reach the variance the SAE explains on held-out positions—403 at layer 0 (SAE 0.848), 317 at layer 5 (0.856), 308 at layer 11 (0.823) and 250 at layer 17 (0.775). A 50-axis basis is five- to eightfold short of the SAE's own reconstruction capacity, so it is the wrong reference for asking what the dictionary can see.

Where SAE directions sit. For each unit-normalized decoder direction we computed the cumulative projection ρ_k onto the rank-k principal subspace, for every k up to the full 1,152 dimensions, against the random-direction expectation $\rho_k = k/d$. SAE directions are systematically concentrated in the leading subspace: at k=100 the median direction has $\rho_k = 0.21$ at layer 0 and 0.33 at layer 11, against 0.087 for a random direction, a two- to fourfold excess. The median direction reaches half of its norm by $k = 247$ (L0), 190 (L5), 178 (L11) and 152 (L17), where a random direction requires 576, and reaches 90% by $k = 664$–764 against 1,037. SAE features are not hidden from the principal subspace; they occupy its high-variance part preferentially.

Low single-axis alignment is not a capacity artefact. The fraction of features whose absolute cosine with at least one of the leading k axes exceeds 0.5 saturates by $k \approx 20$–50 and does not increase thereafter. Even against the *complete* 1,152-dimensional basis it is 0.98% at layer 0, 0.13% at layer 5, 0.09% at layer 11 and 0.26% at layer 17. Offering more axes does not recruit more features, because each feature distributes its norm across many axes rather than concentrating on one. The same conclusion holds across cosine thresholds $\{0.3, 0.5, 0.7, 0.9\}$ (Supplementary Table S13).

Annotation yield per direction. We annotated the leading 50 principal axes with the identical top-20-gene enrichment protocol used for SAE features, in both polarities, since a principal axis is signed whereas a TopK feature is one-sided. Per direction, principal axes are at least as annotatable as features: at layer 0, 98% of axes carry at least one significant term with a mean of 159.5 terms per direction and 2,029 unique terms, against 90%, 64.4 and 1,977 for 100 randomly drawn SAE features; at layer 11 the corresponding numbers are 98%, 178.8 and 2,083 against 93%, 71.3 and 2,045. Biological annotation is therefore not the exclusive property of non-aligned directions.

Taken together, these three measurements locate what the dictionary contributes. It is not access to information that linear decomposition cannot reach—the leading principal subspace contains most of each feature's norm and annotates readily. It is a different *parameterization* of that information: thousands of sparse, one-sided, individually interpretable directions in place of a few hundred dense, signed, polysemantic ones. That is the property the rest of this study relies on, because it is what makes per-feature annotation, per-feature co-activation structure, and per-feature causal intervention possible at all.

How many distinct concepts the atlas contains

Dictionary atoms trained at different layers occupy different vector spaces, so their counts cannot be pooled into a number of concepts held simultaneously in one representation. What can be compared across layers is each feature's gene-level signature. We built a similarity graph over all 82,525 Geneformer features from their top-20 gene sets and partitioned it (Methods; Fig. 6, Supplementary Table S15).

The 82,525 atoms resolve into **18,711 distinct gene-level programs**, that is 4.4 atoms per program. The count is insensitive to the clustering resolution: 18,695 at γ=0.5 and 18,748 at γ=2.0. The distribution is strongly bimodal: 17,690 programs are singletons—features whose gene signature overlaps no other feature's—while 1,021 multi-atom programs absorb the remaining 64,835 atoms, the largest containing 3,221. Redundancy is therefore concentrated in a small number of heavily re-derived programs rather than spread evenly. As exact sets the signatures are nearly all distinct: 82,442 distinct signatures, with only 56 shared by more than one feature.

Programs recur across depth. Of the multi-atom programs, 97.1% appear at two or more layers, and the mean program spans 3.16 layers. Read together with the near-complete representational turnover between adjacent layers, this gives a consistent picture: each layer rebuilds a largely fresh dictionary, but the gene-level programs those dictionaries express are re-derived throughout the stack.

Within a single layer—the only place a features-per-space ratio is defined—the dictionary is 4$\times$ overcomplete (4,608 atoms in 1,152 dimensions, of which 4,598 are alive at layer 11) and resolves 774–2,048 distinct programs depending on depth (1,301 at L0, 814 at L5, 1,252 at L11, 1,360 at L17). The packing is close to the geometric limit: mean absolute decoder coherence is 0.036 at layer 11 against a Welch bound of 0.0255 for 4,608 unit vectors in 1,152 dimensions, so the dictionary is near-optimally spread rather than exploiting an unusually large capacity.

Causal ablation: what a feature is necessary for

Annotation and co-activation establish that features correspond to biology, but not that the model uses them. To test necessity we ablated single features at layer 11: the hidden state was encoded through the SAE, one feature's activation set to zero, the state decoded and substituted, and the forward pass continued to the output logits. Disruption is the mean absolute change in output logits at a set of gene positions, and specificity is that quantity on a gene set divided by the same quantity on all other positions (Fig. 7, Supplementary Table S23).

The design of this test determines what it can show. A feature's annotation is assigned from the genes it most strongly activates on, so evaluating disruption at those same genes asks whether ablating a feature affects the genes that define it—a question whose answer is close to tautological. We therefore partitioned the gene positions of each tested feature into three disjoint sets: genes that are both in the feature's top-20 activating genes and in its annotated term; genes of the annotated term that are *not* among its top-20 (an evaluation set independent of the annotation evidence); and all remaining positions. Each feature additionally received its own null: a random gene set matched to its held-out set in size and in mean-activation decile. Features were drawn from three arms—the most richly annotated features, a uniformly random sample of annotated features, and a uniformly random sample of all alive features—so that the result cannot depend on how features were selected. All 119 tested features come from K562 cells.

Ablation has a large and reproducible effect on the genes a feature activates on. Median specificity on the top-20 genes against all other positions is 35.5× in the richly annotated arm, 64.3× and 62.1× in the two random arms, with over 92% of features exceeding 2× in every arm. Restricted to the genes that are both in the top-20 and in the annotated term, the medians are 37.3×, 56.9× and 66.9×.

On the independent evaluation set the effect disappears. Median specificity on annotated-term genes outside the feature's top-20 is **1.04**, **0.94** and **0.97** in the three arms. The matched random gene sets give 0.89, 0.91 and 0.89, so the held-out sets are statistically indistinguishable from gene sets chosen at random with the same size and activation level: a per-feature Wilcoxon signed-rank test of held-out against matched random gives $p = 0.76$ and $p = 0.34$ in the two unbiased arms, and $p = 0.006$ with a median difference of 0.13 in the richly annotated arm. Fourteen of the 119 features admit no independent evaluation at all, because every gene of their annotated term already lies within their own top-20.

The conclusion is narrower than the ablation magnitudes alone suggest. Single-feature ablation produces a specific, reproducible, gene-level effect, and that effect is confined to the genes on which the feature fires. It does not extend to the remainder of the biological process the feature is annotated with. Feature-level interventions therefore reveal genuine computational structure that component-level ablation misses [3], but the structure they reveal is the feature's own support rather than the pathway it is labelled with.

Cross-model comparison on matched inputs

The two atlases were built from different cells, so any direct comparison of their statistics confounds architecture with input distribution. To separate the two we trained Geneformer dictionaries on the same 3,000 Tabula Sapiens cells used for the scGPT atlas, with identical architecture and hyperparameters, and compared the models at matched relative depth (Supplementary Table S17). Because the scGPT dictionaries were originally trained on all available positions rather than a subsample, we also retrained them under the Geneformer protocol as a control.

On matched inputs there is no uniform architectural advantage. Geneformer reconstructs better at the input end (0.7731 against 0.7258 at relative depth 0), the two are indistinguishable at about a quarter depth (0.7749 against 0.7729), and scGPT is ahead through the second half, reaching 0.8862 against 0.7241 at the output end. The difference between the models is a depth-dependent crossover rather than a constant offset, and it is invisible unless the inputs are matched.

Input distribution accounts for a difference of the same order. Each Geneformer dictionary evaluated on the distribution it was trained on reconstructs the CRISPRi-derived control cohort 4.7–7.8 percentage points better than the Tabula Sapiens dictionary reconstructs Tabula Sapiens—0.8444 against 0.7731 at layer 0, for example—so a difference of this size between two models is what differing inputs alone can produce. Both dictionaries sit close to the packing limit for their dimensionality: mean absolute decoder coherence is 0.030–0.036 for Geneformer and 0.039–0.047 for scGPT, against Welch bounds of 0.0255 and 0.0383 respectively.

Robustness to training choices

Architecture. Grid-training at layer 11 over expansion ratio $\{2\times, 4\times, 8\times\}$ and sparsity $k \in \{16, 32, 64\}$ gives variance explained rising monotonically with both, from 0.706 to 0.884, with the reported configuration in the middle at 0.819 (Supplementary Table S12). Module count stays between 6 and 12 across the entire grid and mean decoder coherence between 0.033 and 0.042, so the reported structure is not a property of one corner of the hyperparameter space.

Random seed. The atlas dictionaries were trained with seed 42. Retraining at layers 0 and 11 under six runs each—five differing only in initialisation and batch order, one with a different training subsample—separates what is a property of the model from what is a property of the run (Supplementary Table S16).

Everything the atlas reports in aggregate is reproducible. Variance explained is 0.8454 ± 0.0003 at layer 0 and 0.8200 ± 0.0003 at layer 11, the annotation rate is 0.935 ± 0.009 and 0.933 ± 0.010, module counts are 7.2 ± 0.4 and 8.3 ± 1.0, and dead-feature counts are 2 ± 1 and 32 ± 4.

Individual dictionary atoms are not. The mean best decoder cosine between runs is 0.282 at layer 0 and 0.294 at layer 11, and fewer than 1% of atoms have a counterpart above 0.9 in another run. The reference for these numbers is what a random direction achieves: the best match a random unit direction finds among 4,608 directions in 1,152 dimensions is 0.113 ± 0.008, and no random direction reaches 0.9. Cross-seed agreement is therefore well above chance and far below reproducibility. Independent runs learn different bases that explain the same variance and carry the same annotation. This bears on how the released atlases should be read: their aggregate structure is a property of the model, while the identity of any individual catalogued feature is partly a property of the training run—consistent with the gene-level analysis above, in which the recurring unit across layers is the program rather than the atom.

Features without ontology annotation

Between 41% and 55% of features carry no significant ontology annotation, and it matters whether these are structure the databases do not cover or noise. Clustering unannotated features by top-20 gene-set similarity places only 2–3% into standalone gene-set clusters, among them ribosomal protein programs and mitochondrial complex assembly (Supplementary Table S25).

Two tests separate them from annotated features (Supplementary Table S24). Cell-type specificity does not: at layers $\{0, 5, 11, 17\}$, 89.4–95.0% of unannotated features are enriched for at least one cell type against 94.5–95.8% of annotated ones, and a matched random-feature null sits at 91.9–94.7%, so essentially every feature at these layers is cell-type selective and the test has no discriminating power. Perturbation responsiveness does: at layer 11, 2.32% of unannotated features appear in a perturbation response list against 0.15% of annotated features, a fifteenfold difference, with unannotated features more responsive at layers 0 and 5 and indistinguishable at layer 17.

A cautious reading is that some unannotated features carry perturbation-response signal the ontology databases do not describe. Being non-random is not the same as being biologically meaningful, however: a feature that is cell-type selective may equally be tracking donor or assay structure, which is why the batch analysis above is reported alongside.

What the perturbation data can support

The preceding sections establish that features are biologically annotated, modularly organized and causally specific. The question that remains is whether they encode regulatory logic: when a transcription factor is knocked down, do the features that respond correspond to that factor's known targets?

A recovery rate answers that question only against a reference. Before measuring the atlas we therefore asked how much curated regulatory structure the assay can express at all. The CRISPRi resource measures 6,546 genes. Of the 73 perturbed transcription factors with at least 50 cells, only nine have five or more TRRUST targets among the measured genes, three have ten or more, and one has twenty or more; the median perturbed factor has exactly **one** measurable target (Fig. 8b, Supplementary Table S18). DoRothEA, whose regulons are larger, does better—13 factors with five or more measured targets at the A+B+C confidence tier, median 15—but the constraint is severe under every reference. For most of the panel, target-specific recovery is not merely difficult; it is undefined, for any method.

A recoverability benchmark

We therefore evaluated the SAE alongside methods that span the range of what is possible on these data, each emitting a ranked list of 100 putative targets per factor and each scored by the same hypergeometric enrichment with Benjamini–Hochberg correction across the panel and the measured genes as the universe (Fig. 8a, Supplementary Table S19).

Differential expression of the knockdown itself is the perturbation-aware reference: it uses the same cells and the actual intervention, and therefore bounds what any observational method can achieve here. It recovers curated targets for 1/9 factors against TRRUST (11.1%), 3/13 against DoRothEA A+B+C (23.1%) and 8/18 against DoRothEA at all confidence levels (44.4%). GENIE3 reaches 11.1%, 7.7% and 0.0% under the three references, GRNBoost2 11.1%, 7.7% and 5.6%, Pearson co-expression 22.2%, 15.4% and 11.1%, and Spearman 22.2%, 15.4% and 5.6%. Contextual gene embeddings taken directly from the model, without any sparse decomposition, reach 11.1%, 7.7% and 0.0%, and the SAE feature response recovers targets for no factor under any reference (0/9, 0/13, 0/18). Size-matched random gene sets recover nothing either, confirming that the test is calibrated.

Two things follow. On the best-powered reference no observational method approaches the intervention: 44.4% against 0–11.1%. Under the sparser references the ceiling is itself so low—11.1% under TRRUST—that co-expression matches or exceeds it, which is what a median of one measurable target per factor produces: the reference is too thin to rank methods at all. The choice of universe matters for this calibration: scoring against a nominal 20,000-gene universe rather than the 6,546 measured genes makes random gene sets appear significant for 16.7% of factors, because predictions can only be drawn from what is measured.

Perturbation responses are detectable but not transcription-factor specific

We mapped perturbation responses at layer 11 using up to 200 CRISPRi cells per target against 600 non-targeting cells of the same line, encoding every cell through the atlas dictionary (Methods). The panel comprises 20 perturbed transcription factors, 18 of them evaluable, and as a negative control ten perturbations of genes that are not transcription factors under any reference.

The unit of replication. A perturbation is applied per cell, but a per-position statistic treats each gene position as an observation. The intraclass correlation of feature activation across positions within a cell is 0.000298, which at the observed 2,020 positions per cell gives a design effect of 1.61: the effective sample size is about 62% of the nominal position count. The correction is worth making and we make it throughout—significance is assessed over cells—but its magnitude is bounded, because TopK sparsity leaves activations at different positions of one cell close to independent.

Responses are weak and non-specific. Thresholding on effect size alone—a shift exceeding half a control standard deviation, with no correction for testing 4,608 features—the panel yields a mean of 3.20 responding features per factor, with 15 of 20 factors showing at least one. Correcting for multiple testing across features collapses this to two of twenty, so most of what a per-feature threshold registers does not survive the number of features being tested. Under that correction the like-for-like comparison of the two units of replication gives 1.60 responding features per factor pooling positions and 1.90 assessing over cells (Supplementary Table S20). The ten non-transcription-factor controls yield *more* responding features than the transcription factors (2.30 versus 1.90), so what the features register is not specific to transcription-factor perturbation.

Where the sensitivity ends. The clearest measurement of the limit compares the two readouts on identical cells. Differential expression detects the knocked-down gene itself in 15 of 16 targets, at FDR < 0.05 with a median Cohen's d of -1.19 (range -2.76 to -0.49). The SAE feature response detects it in 1 of 20. The perturbation is therefore unambiguously present in the expression profile of the same cells and is not registered by the feature responses. The features are not insensitive because the data are thin; they track global cell-state shifts rather than the gene-level consequences of the intervention. This is the specific sense in which detection is limited, and it is a property of what the dictionary encodes.

Sample size. Restricting to the 14 factors with at least 100 cells, so that sample size is not confounded with which factors are available, the responding-feature count is flat across a tenfold increase in cells (0.56 at n=10, 0.65 at n=20, 0.65 at n=50, 0.61 at n=100). Detection of the knockdown rises from 5.0% to 7.1% and saturates. Nominal target enrichment improves from n=10 to n=50 (7.7% to 27.5%) and then plateaus, the gain saturates well before the effect becomes substantial, and under correction across the panel no sample size yields a significant factor (Supplementary Table S21).

Widening the training distribution. A dictionary trained on the same control cells pooled with 3,000 Tabula Sapiens cells from three primary tissues gives 2.55 responding features per factor against 1.90 for the atlas dictionary, with the same non-transcription-factor controls at 2.30 and detection of the knockdown at 0 of 20. Diversifying the training distribution beyond immortalized lines does not make regulatory structure recoverable.

Replication across cell lines and independent datasets

The measurement was repeated in three further cell lines from the same resource, with perturbed and non-targeting cells drawn from the same line in every case, and on two perturbation datasets generated by other groups with other protocols and substantially larger gene panels (Fig. 9c, Supplementary Table S22).

Across K562, RPE1, Jurkat and HepG2 the pattern is the same. Pooling the four lines, **1 of 66** evaluable transcription factors shows target enrichment at FDR < 0.05 (1.5%): none of 18 in K562, none of 16 in RPE1, one of 16 in Jurkat and none of 16 in HepG2. The knocked-down gene itself is detected in 2 of 68 factors across the four lines. Responding-feature counts are low throughout (means of 0.06–1.90 per factor) and do not separate transcription factors from the non-transcription-factor controls, which average 0.17–2.30 in the same lines. The design effect is 1.61–1.73 everywhere, so the correction for the unit of replication behaves consistently across backgrounds. RPE1 is a non-transformed retinal pigment epithelial line and HepG2 a hepatocyte line, so the result is not confined to the leukaemic backgrounds in which these models are most often probed.

Both external datasets measure far more of the transcriptome than the primary resource, and the difference changes the result. In THP-1 monocytes 18,649 genes are measured and the median perturbed factor has 19.5 curated TRRUST targets among them; in the K562 Perturb-seq data 33,694 genes are measured. The Perturb-seq screen is a CRISPR *activation* experiment, so it also tests the question under gene induction rather than repression: the median perturbed factor is induced with a Cohen's d of $+0.98$ and yields 405 differentially expressed genes.

On these data the features do recover targets. Against DoRothEA the SAE reaches 20–22% of factors in the THP-1 data and 23–25% in the Perturb-seq data, where on the primary panel it reached none. Against TRRUST it reaches none in either, while differential expression on the same cells reaches 18% and 30%.

Whether that recovery is transcription-factor-specific requires a control, and we ran one: ten perturbations of genes absent from every reference network, put through the identical pipeline, with their predicted sets tested against the *same* regulons as the transcription factors.

Against the high-confidence DoRothEA subset, a factor's own predicted set is enriched for its own regulon in 3 of 13 cases (23%). Of the nine control perturbations that produced any prediction, one is enriched for at least one factor's regulon (11%), giving 1 significant test in 117 factor-by-regulon comparisons. The difference is in the direction specificity would predict, but a 13-factor panel cannot establish it: Fisher's exact test on 3 of 13 against 1 of 9 gives $p = 0.62$.

Against the union of all confidence levels the comparison is decisive in the other direction. The matching factors reach 4 of 16 (25%) while 8 of the 9 control perturbations enrich for at least one regulon ($p = 0.004$, controls higher). Those regulons have a median of 402 measured genes and reach 9,321, and a broad expression signature enriches for them regardless of which gene was perturbed. The all-confidence tier does not measure regulatory recovery and we do not read it as such.

Two conclusions follow. The absence of detectable transcription-factor specificity in the primary CRISPRi panel is substantially a property of that assay: with a median of one measurable target per factor, neither these features nor differential expression on the same cells nor established network inference can demonstrate specificity there. On perturbation data that measure the transcriptome the features do produce enrichment for curated regulons, at a rate above the non-regulatory controls under the high-confidence reference and below them under the permissive one. The high-confidence comparison is the interpretable one and it is under-powered: three factors against one control gene, on a panel of thirteen. What can be stated is that these data do not reproduce the flat null obtained on the primary panel, and that establishing whether a small factor-specific signal exists would require a substantially larger panel than any single published screen currently supports.

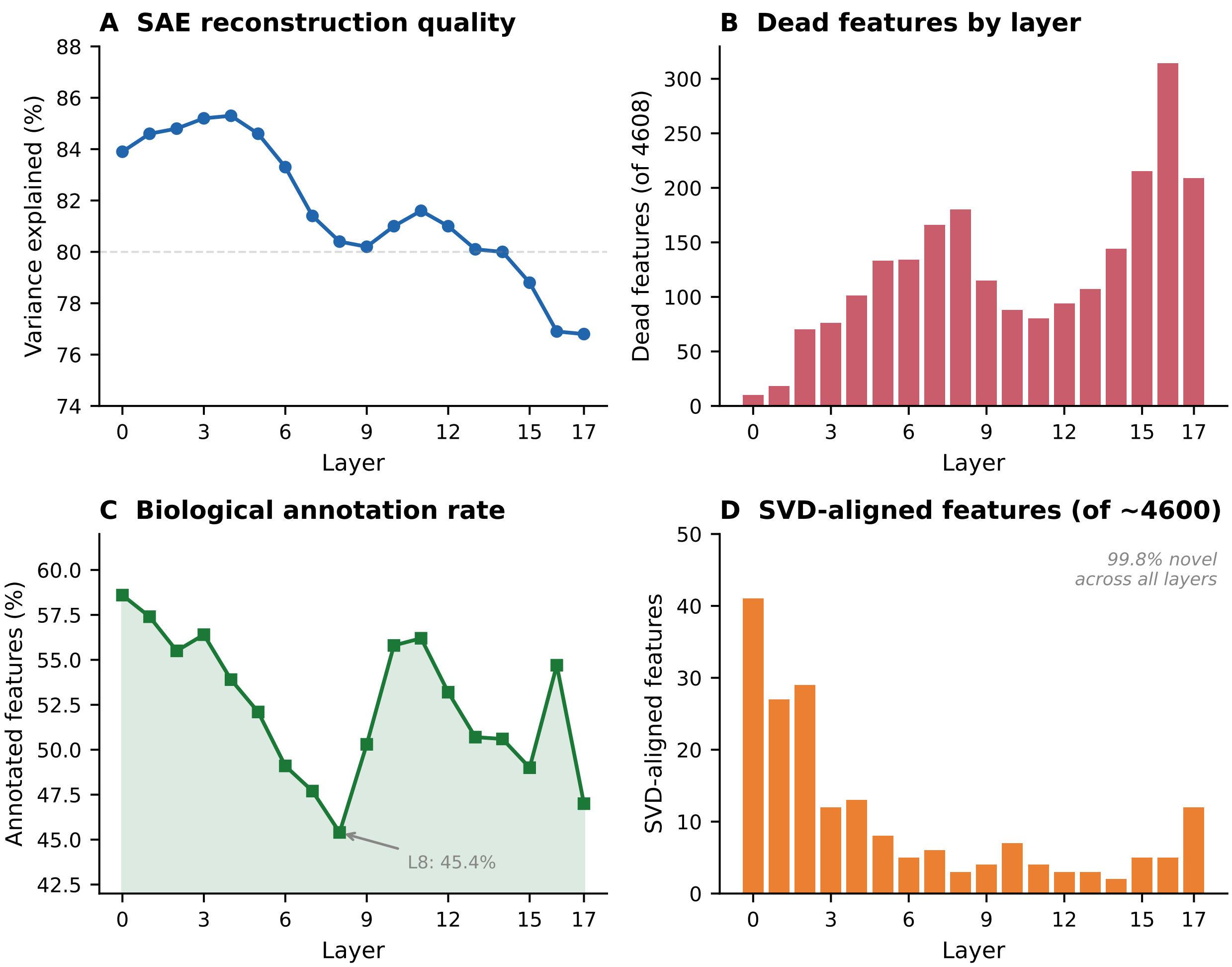

Figure 1: **Sparse autoencoder feature atlas of Geneformer V2-316M.** TopK sparse autoencoders trained on all 18 layers yield 82,525 alive features. Reconstruction quality declines with depth while dead features increase.

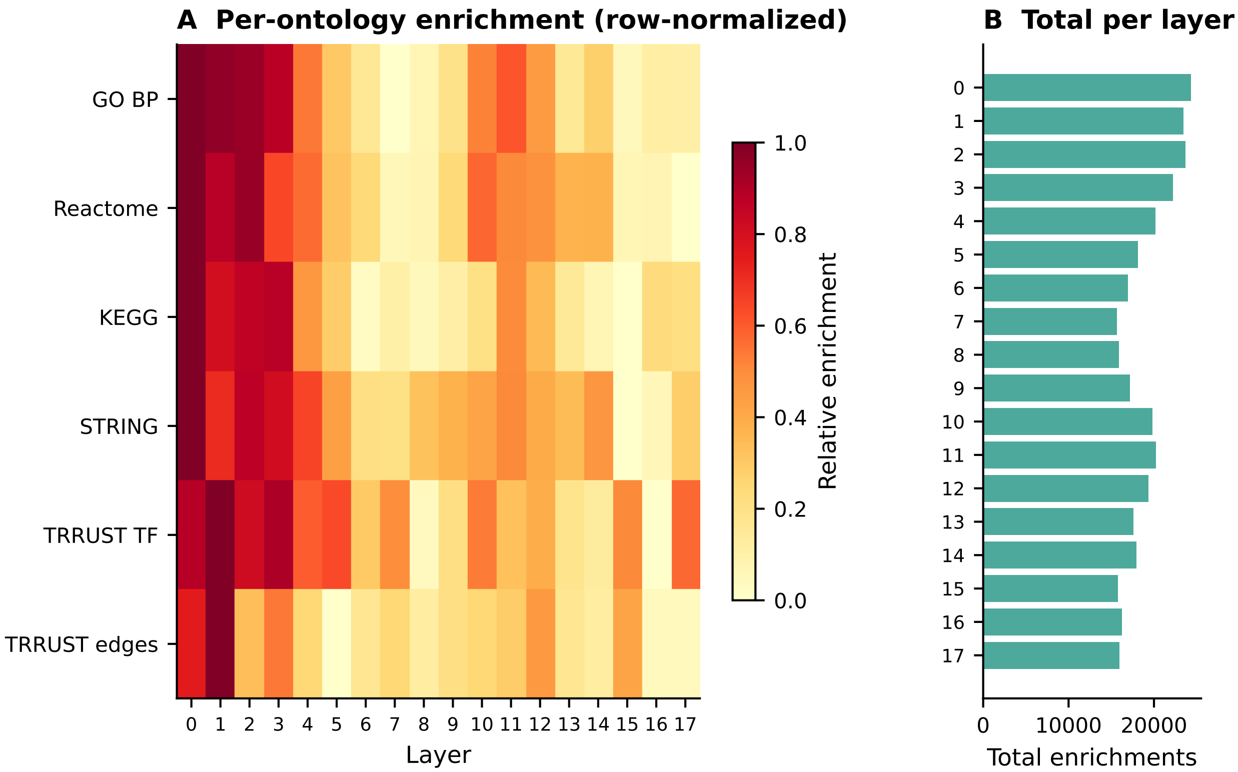

Figure 2: **Biological annotation follows a U-shaped profile across layers.** Early layers encode molecular programs, middle layers develop more abstract representations, later layers re-specialize before becoming prediction-focused. Enrichment uses Gene Ontology, KEGG (Kanehisa Laboratories), Reactome, STRING and TRRUST gene sets.

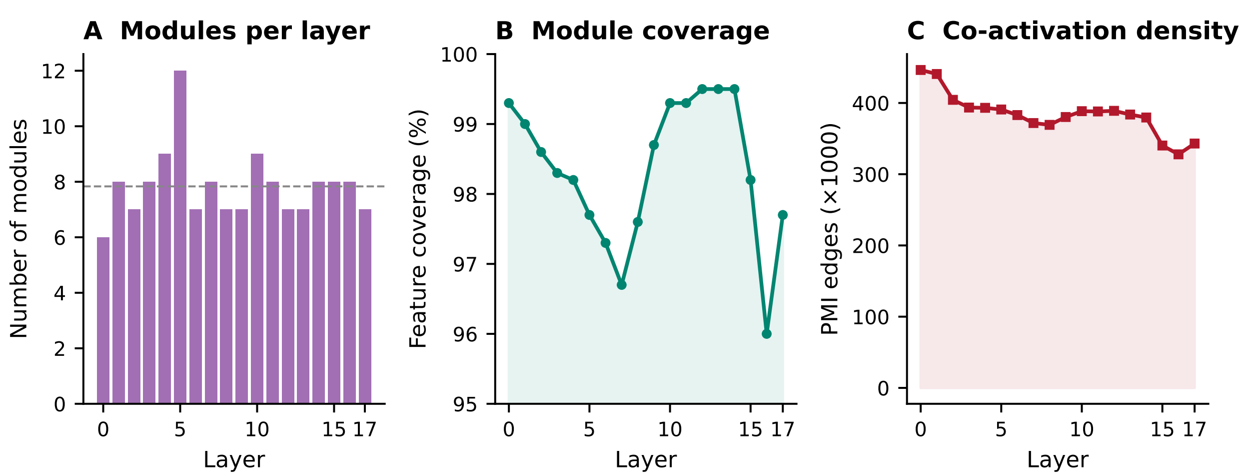

Figure 3: **Co-activation module structure across layers.** (A) Modules per layer. (B) Fraction of alive features covered by a module. (C) Co-activation edge density.

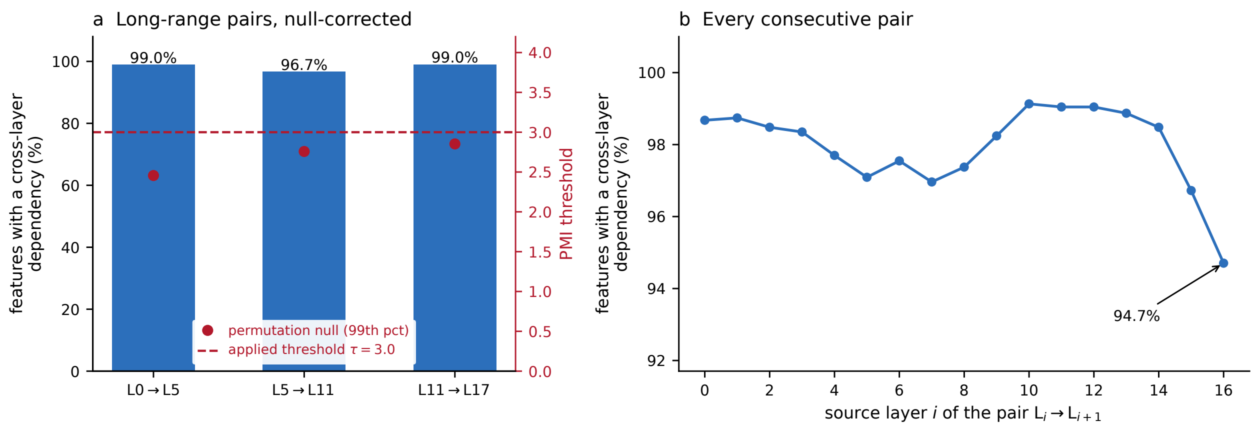

Figure 4: **Cross-layer information flow is continuous.** (a) Highway fraction for the three long-range layer pairs after permutation-null correction; the 99th-percentile null threshold sits below the applied threshold for every pair, so the correction does not shift the fractions. (b) The same quantity for all 17 consecutive layer pairs, with no sharp drop at any transition.

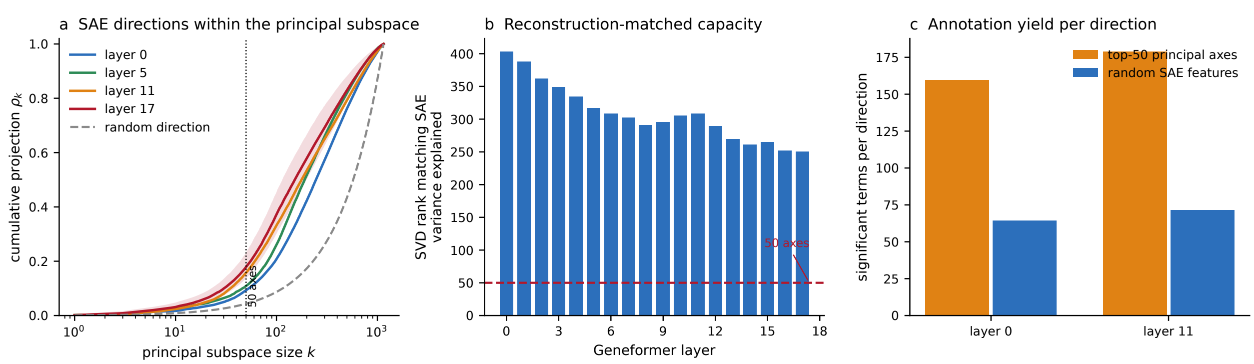

Figure 5: **The dictionary re-parameterizes the principal subspace rather than reaching past it.** (a) Cumulative projection ρ_k of each decoder direction onto the rank-k principal subspace, median over features, against the random-direction expectation k/d; the shaded band is the interquartile range at layer 17 and the dotted line marks the 50-axis basis. (b) Truncated-SVD rank required to match the SAE's own variance explained, per layer. (c) Significant enrichment terms per direction when the leading 50 principal axes are annotated with the protocol used for SAE features. Enrichment uses Gene Ontology, KEGG (Kanehisa Laboratories) and Reactome gene sets.

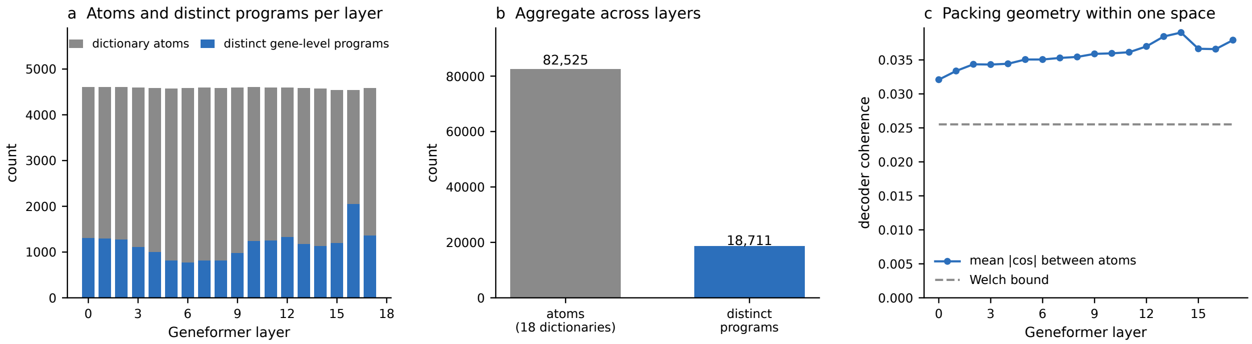

Figure 6: **Dictionary atoms and the distinct programs they express.** (a) Atoms and distinct gene-level programs per layer. (b) Aggregate across all 18 dictionaries. (c) Mean absolute coherence between unit-normalized decoder atoms against the Welch bound, the smallest achievable maximum coherence for that many unit vectors in 1,152 dimensions.

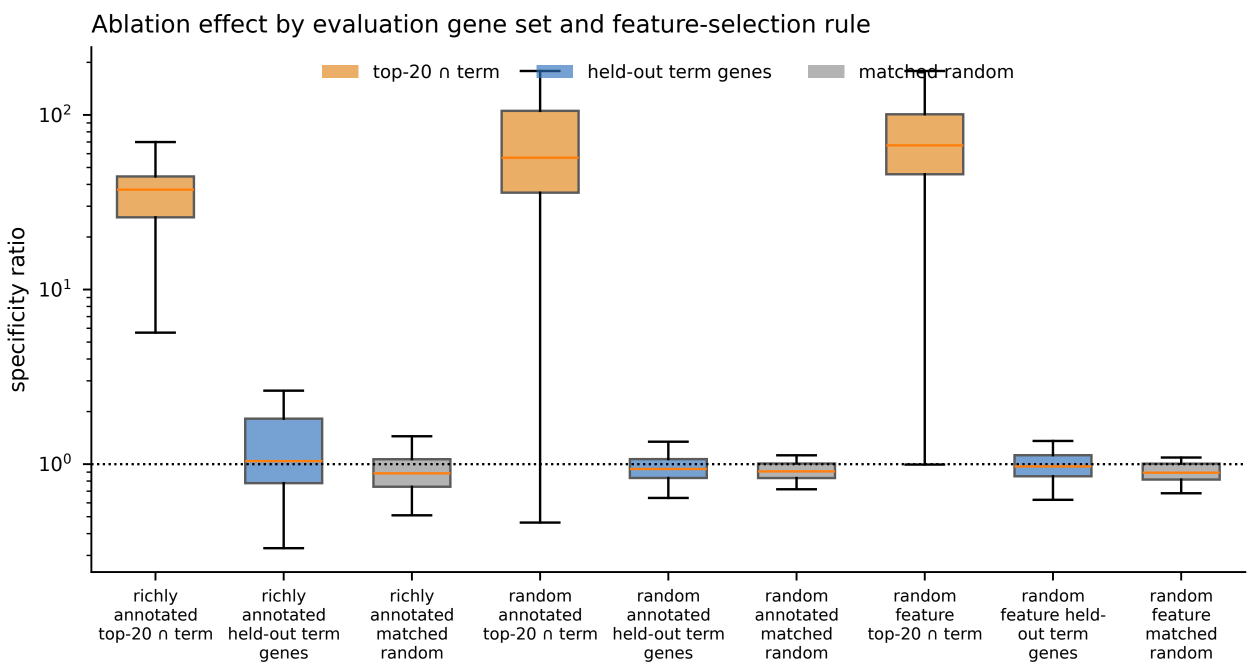

Figure 7: **Causal feature ablation with an independent evaluation set.** Distribution of specificity ratios across features, for three feature-selection rules and three evaluation gene sets: the feature's top-20 genes intersected with its annotated term; the annotated term's genes excluding that top-20; and a size-matched random gene set stratified on expression rank and detection frequency, drawn per feature. The dotted line marks a ratio of one.

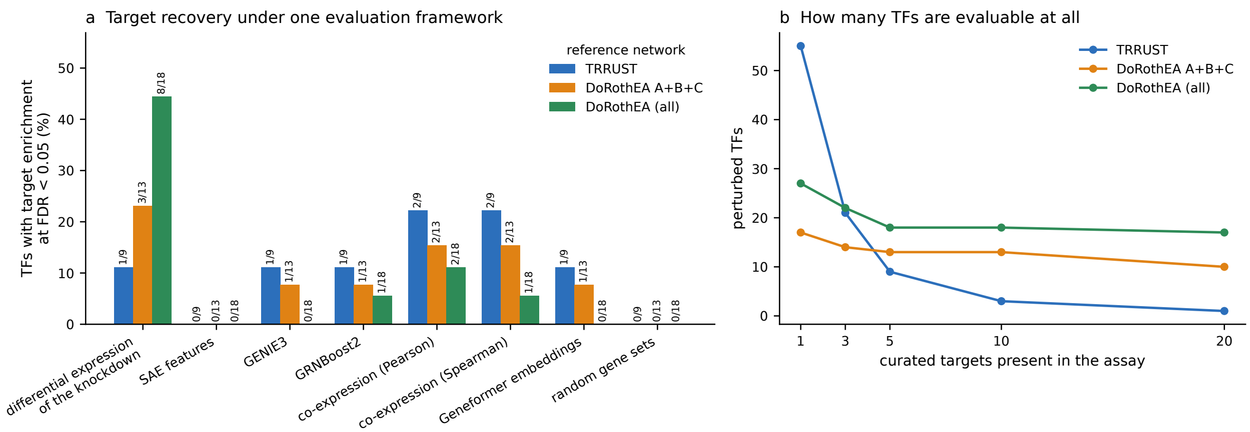

Figure 8: **What these data can support.** (a) Fraction of transcription factors whose predicted target set is enriched for its curated targets, at matched prediction-set size and under one hypergeometric framework, for every method. (b) How many perturbed transcription factors have enough curated targets among the measured genes to be evaluable at all.

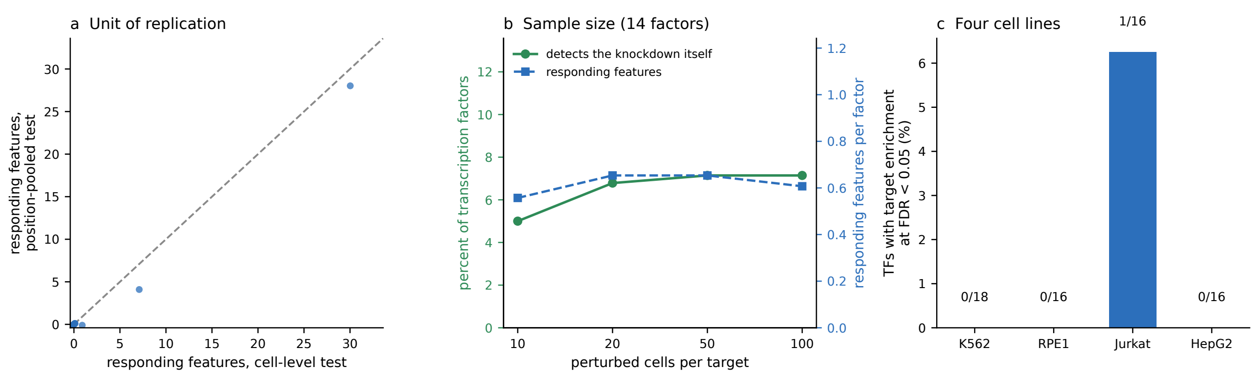

Figure 9: **Unit of replication, sample size, and replication across cell lines.** (a) Responding features per target under a cell-level test against a position-pooled test on the same cells; the dashed line is equality. (b) Detection of the knocked-down gene itself, and the number of responding features, as a function of perturbed cells per target, restricted to the 14 factors with at least 100 cells so that sample size is not confounded with which factors are available. (c) Fraction of evaluable transcription factors with target enrichment in each cell line, with perturbed and control cells drawn from the same line.

3 Discussion

We have presented a systematic application of sparse autoencoders to single-cell foundation models, producing feature atlases for two architecturally distinct models: 82,525 features across 18 layers of Geneformer V2-316M and 24,527 features across 12 layers of scGPT whole-human. Both atlases reveal models that have organized biological knowledge into modular, interconnected internal representations. Against that organization stands a specific negative result: the evidence that they carry transcription-factor-to-target regulatory logic is bounded above by what the available perturbation assays can measure, and on the assay this atlas was built from that bound is close to zero.

Superposition is real but is not invisibility. Comparing thousands of dictionary atoms with a handful of principal axes cannot establish that a model's structure is inaccessible to linear decomposition, because the two sides of that comparison differ in capacity by two orders of magnitude. Under a matched comparison the picture is sharper and more interesting. SAE directions are not principal axes—even against the complete 1,152-dimensional basis, under 1% of features align with any single axis above a cosine of 0.5—but neither are they hidden from the principal subspace: the median direction places half of its norm in the leading 150–250 components, well ahead of the random-direction reference, and truncated SVD needs rank 250–403 to match the SAE's own reconstruction quality. Nor is biological annotation the exclusive property of non-aligned directions: annotated with an identical protocol, the leading 50 principal axes yield at least as many enriched terms per direction as randomly drawn SAE features. What the dictionary provides is not access to otherwise unreachable information but a *different parameterization* of it—many sparse, one-sided, individually interpretable directions in place of a few dense, signed, polysemantic ones. That is a real and useful property, and it is the one worth claiming.

What a feature count does and does not mean. Dictionary atoms trained at different layers occupy different vector spaces and cannot be summed into a count of concepts coexisting in one representation. Within a single layer the dictionary is 4× overcomplete and resolves roughly 1,250 distinct gene-level programs from 4,608 atoms; in aggregate the 18 dictionaries hold 82,525 atoms that resolve 18,711 distinct programs, with 97% of multi-atom programs recurring in two or more layers. The interesting quantity is not a compression ratio but this pattern of re-derivation: each layer rebuilds a largely fresh dictionary, yet the gene-level programs those dictionaries express recur throughout the stack.

Hierarchical biological abstraction across layers. The U-shaped annotation profile, the shift in module themes from molecular machinery to integrative programs, and the near-complete representational turnover between adjacent layers together describe hierarchical biological abstraction. This parallels findings in large language models, where early layers handle token-level processing and later layers more abstract semantic computation, but here the tokens are genes and the semantics are biological programs.

Feature-level structure versus component-level nulls. Ablating a single feature produces a large, reproducible and gene-specific change in the model's output where ablation of whole attention heads or MLP layers produces null behavioural effects [3]. Computation is therefore organized around directions in the residual stream rather than around architectural components, which is why component-level interventions missed it. The scope of that effect is narrow, however: it falls on the genes the feature fires on and not on the remainder of the biological process the feature is annotated with, which is indistinguishable from a size- and activation-matched random gene set. A feature is a handle on its own support, not on the pathway its label names—a distinction that matters for anyone intending to use SAE features as intervention targets.

The co-expression–regulation boundary, and where it actually lies. Our results extend to the residual stream a boundary that has now been located by several independent probes of the same models. Attention-derived edge scores add no incremental predictive value over gene-level features for perturbation-target prediction [3]. Causal circuit tracing recovers a dense, biologically coherent computational graph whose edges do not correspond to directed regulatory relationships [21], and exhaustive mapping of that graph finds massive redundancy rather than identifiable regulatory pathways [22]. Direct causal intervention on gene representations in scGPT does not recover regulatory targets [23], and adversarial validation of in-silico perturbation profiles shows the same for predicted perturbation responses [24]. Analyses of the representation geometry reach the same place from a different direction: the structure these models learn is organized by cell state and pathway membership [25, 26], and although residual-stream geometry does carry an incremental gene-regulatory signal, that signal is fragile and does not survive stricter cross-validation [27]. Sparse dictionaries are one more probe, applied to the one representation the others did not decompose, and they find what the others found.

No finite set of probes can establish that regulatory information is absent from a model, because the proposition that it resides in some representation not yet examined is not refutable by examination. What can be done is to state which representations have been examined, with which instruments, and what was found. Attention weights, individual components, the causal graph connecting them, gene-embedding geometry, in-silico perturbation responses, and now a sparse decomposition of the residual stream have each been examined; none yields transcription factor-to-target structure above what the underlying measurements support on the assays where each was applied. That is the sense in which we take the finding to be a property of these models rather than of any one instrument.

The important qualification is what that boundary can be attributed to, and here the evidence is two-sided in a way that the primary dataset alone could not reveal.

A specificity rate is uninterpretable without knowing how much of the regulatory relationship is recoverable from the same measurements. On the CRISPRi resource the atlas was built from, that ceiling is very low: the assay measures 6,546 genes and the median perturbed transcription factor has a single curated target among them. Differential expression computed directly on the knockdown recovers curated targets for a minority of factors, established network inference recovers no more, and the sparse features recover targets for 1 of 66 factors across four cell lines. The correct reading of that panel is not that the models lack regulatory structure but that the measurement cannot decide the question.

On perturbation data that measure the transcriptome the flat null does not reproduce. In a CRISPR-activation screen with 33,694 genes measured and strong induction, a factor's predicted target set is enriched for its own high-confidence regulon in 3 of 13 cases, against 1 of 9 perturbations of genes absent from every reference network. That is the direction specificity predicts, and the panel is too small to establish it ($p = 0.62$). Against low-confidence regulons of several hundred genes the comparison reverses outright—8 of 9 controls enrich—so that tier measures breadth of the predicted set rather than regulatory recovery.

What this study can support is therefore bounded on both sides. It is not the absence of regulatory information: the assay the atlas was built on cannot detect such information if it is present, and on data that can, the null does not hold. Neither is it the presence of a factor-specific signal: the one comparison capable of demonstrating that rests on thirteen factors in a single screen and does not reach significance. Settling the question needs a perturbation panel substantially larger than any single published screen provides. What we can state without qualification is that an interpretability result reported on a 6,546-gene assay is measuring the assay as much as the model.

Where the unit of analysis matters, and by how much. A perturbation is applied to a cell, but a per-position statistic treats every gene position in that cell as a separate observation. Whether this matters depends on how strongly feature activation is correlated within a cell, which is an empirical question we can answer rather than assume. The intraclass correlation of SAE feature activation across positions of the same cell is 2.98×10^{-4}, which for the observed 2,020 positions per cell gives a design effect of 1.61, that is an effective sample size of 62% of the nominal count. The effective sample size is therefore a modest fraction of the nominal position count rather than smaller by three orders of magnitude: sparse feature activations at different gene positions of one cell are close to independent, because the sparsity pattern dominates the within-cell variance.

The correction is nonetheless the right one to make, and we make it: significance is assessed with the cell as the unit of replication throughout, holding the effect-size definition fixed at the position-level scale so that the cell-level and position-level results differ only in what is counted as an independent observation. We recommend the cell-level formulation as the default for any interpretability analysis that tests per-position statistics against a per-cell intervention, while noting that its practical effect here is bounded and measurable.

Implications for foundation model training. Current scFM training objectives may inherently bias representations toward co-expression. Learning causal regulatory relationships would require training signals that distinguish cause from correlation, such as perturbation-prediction objectives incorporated during pre-training. Equally, evaluating whether a model has learned such relationships requires assays that measure the target genes: a genome-scale perturbation atlas with a restricted expression panel bounds what any interpretability method can demonstrate, however good the method is.

SAEs as a general interpretability framework for biological models. Independent of the regulatory question, this work establishes SAEs as a productive framework for scFM interpretability. Applying an identical pipeline to two architecturally distinct models, with qualitatively consistent results, demonstrates that these tools transfer to any transformer-based biological model.

What would settle it. Regulatory structure in these representations should be assessed on perturbation atlases that measure the whole transcriptome, against high-confidence regulons, with perturbations of non-regulatory genes as a matched control, and with the recoverable ceiling computed on the same cells. We apply that design here, and its limitation is scale rather than construction: thirteen evaluable factors in one screen cannot separate a modest specific signal from none. A panel several times larger, or several screens combined under a common evaluation, would. Applied instead to a 6,546-gene assay, any interpretability method returns a null whose cause cannot be identified, and we would encourage reporting the evaluability of the assay alongside any claim about what a model encodes.

Limitations. Several bounds apply to what we report. The regulatory analysis is limited by the gene panel of the perturbation assay, as quantified above; a perturbation resource measuring the full transcriptome would raise the ceiling against which any method is judged. TRRUST and

DoRothEA are themselves incomplete and literature-biased; a ChIP-seq-derived catalog would give an orthogonal view we have not pursued. Our SAEs use one point in a wide architecture space, and although a targeted expansion-by-sparsity ablation and a multi-seed stability analysis both leave the conclusions intact, the space is not exhausted. Causal patching tests single-feature ablation; combinatorial effects remain unexplored. Our SAEs are trained on unperturbed activations, which may bias the learned dictionary toward cell-state and co-expression programs; training on pooled perturbed and control activations with perturbation identity as an auxiliary signal is the most direct way to test this and is beyond the scope of the present analysis. Finally, donor and assay identity are not recorded in the cached extraction metadata for the Tabula Sapiens arm, so the batch analysis rests on tissue as a coarse proxy.

4 Methods

Models and data

Geneformer. We used Geneformer V2-316M [1] (18 transformer layers, 1,152 hidden dimensions, 18 attention heads per layer) from HuggingFace (`ctheodoris/Geneformer`, subfolder `Geneformer-V2-316M`). Perturbation data come from the Replogle genome-scale CRISPRi resource [12] as a single processed matrix spanning four cell lines (K562, RPE1, Jurkat, HepG2; 643,413 cells, 6,546 measured genes, 2,023 perturbation targets plus non-targeting controls).

Control cohort. The atlas was built from 2,000 non-targeting control cells sampled at random (seed 42) from the 39,165 non-targeting cells in the resource. Because the control condition is defined by perturbation label, this cohort spans all four lines: 591 Jurkat, 581 K562, 567 RPE1 and 261 HepG2 cells (mean 2,028 genes per cell, 4,056,351 total token positions). The atlas dictionaries are therefore learned from a multi-cell-line control population, and we refer to them throughout as the multi-line control atlas rather than attributing them to any single line. Cohort composition for every analysis is recorded in Supplementary Table S1. For the multi-tissue comparison we additionally used 3,000 cells from Tabula Sapiens [28] (1,000 immune cells across 43 cell types, 1,000 kidney cells, 1,000 lung cells), yielding 5,623,164 token positions.

Cell-line assignment in downstream analyses. Analyses that compare perturbed against control cells draw both from the same line, so that a difference between the two cannot reflect a difference in cell-line composition. Causal patching is likewise run within a single stated line. Where an analysis uses the atlas dictionary on cells of one line, this is a deliberate transfer setting and is labelled as such.

scGPT. We used scGPT whole-human [2] (12 transformer layers, 512 hidden dimensions, 8 attention heads per layer), trained on approximately 33 million single-cell transcriptomic profiles. scGPT ingests each cell as a pair of parallel token sequences: gene-identity tokens drawn from a vocabulary of approximately 60,700 genes, and *binned* expression-value tokens (log1p-normalized counts discretized into 51 bins by default). Genes are sorted by expression level (descending) and padded or truncated to a maximum sequence length of 1,200. Pre-training uses an attention-masking scheme (“gene prompt” and “value prompt” modes) that enables generative prediction of expression values conditional on gene identity, which we refer to throughout as a generative or causal-attention objective to distinguish it from Geneformer’s bidirectional masked-token objective. Activations were extracted from the same 3,000 Tabula Sapiens cells, yielding 3,561,832 token positions per layer.

Input handling for each model. The two models differ at the first layer, and this difference propagates through all downstream analyses. For Geneformer the input is deterministic given a cell: the per-gene expression vector is sorted, each gene receives a rank token, and no continuous magnitude is passed to the model. For scGPT both the gene-identity token and its bin index are embedded and summed before the transformer stack, so the model has direct access to a coarse magnitude signal. Residual-stream activations therefore capture, at every layer, a representation whose only expression signal is gene *rank* for Geneformer, and one that includes both rank-ordered gene identity and a binned magnitude for scGPT.

Activation extraction

For both models we performed forward passes with PyTorch hooks registered at each transformer layer's output (after the residual connection), collecting hidden states at every gene position. Activations were stored as memory-mapped NumPy arrays (float32). Extraction used Apple Silicon MPS acceleration. For Geneformer (d=1,152) this is 336.4 GB for the 18 control-cohort layers (18.7 GB per layer) and 103.6 GB for four Tabula Sapiens layers; for scGPT (d=512), approximately 82 GB for 12 layers.

For the perturbation analyses, where the quantity of interest is a per-cell summary rather than the raw residual stream, we instead hooked the target layer directly, encoded the captured hidden state through the layer's SAE on the accelerator, and reduced to per-cell statistics (mean activation, firing rate, and mean square of each feature over that cell's gene positions) before transfer. Cells were batched by sequence length. This keeps the cell as the retained unit of observation and makes the per-cell feature matrices small enough to resample freely.

TopK sparse autoencoder architecture and training

We used TopK SAEs [11] parameterized by input dimension d. The encoder maps $\mathbf{x} \in \mathbb{R}^d$ (centered by subtracting the training-set mean $\boldsymbol{\mu}$) through a linear projection to a pre-activation vector in $\mathbb{R}^{4d}$, followed by TopK sparsification retaining only the k=32 largest activations, to which a ReLU is applied. The decoder reconstructs via $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{h}_{\text{sparse}} + \mathbf{b}_{\text{dec}}$, where W_{dec} columns are re-normalized to unit norm after each gradient step. Training minimized reconstruction MSE; sparsity is enforced structurally by the TopK operation rather than by an L_1 penalty.

For Geneformer (d=1,152): 4,608 features per layer, trained on a 1M-position subsample per layer. For scGPT (d=512): 2,048 features per layer, trained on all 3,561,832 positions per layer. Both used Adam, learning rate 3×10^{-4}, batch size 4,096, and 5 epochs. The atlas SAEs were trained with random seed 42, which also fixes the training subsample; the sensitivity of every reported quantity to this choice is characterized in the seed-stability analysis below.

Feature analysis and ontology annotation

For each alive feature (activation frequency > 0 on 100K held-out positions) we identified the top 20 genes by mean activation magnitude. Each feature was tested for enrichment against Gene Ontology Biological Process [13], KEGG pathways [14], Reactome pathways [15], STRING protein–protein interactions [16], and TRRUST TF–target relationships [17]. TF–target priors from DoRothEA [29] were used at the A+B+C confidence subset and at the union of all confidence levels. Enrichment was assessed with a one-sided Fisher's exact test and Benjamini–Hochberg FDR at $\alpha = 0.05$, with term sizes restricted to 5–500 genes.

Comparison against the principal subspace

At each layer we computed the exact eigendecomposition of the activation covariance from a 500,000-position sample (40 evenly spaced contiguous blocks), giving all d principal directions and their eigenvalues. Three comparisons follow.

Reconstruction-matched capacity. The truncated-SVD rank k^* at which cumulative eigenvalue mass equals the SAE's variance explained on held-out positions.

Cumulative projection. For each unit-normalized decoder direction $\mathbf{d}_f$ we computed $\rho_k = \|P_k \mathbf{d}_f\|^2 = \sum_{i \le k} (\mathbf{d}_f \cdot \mathbf{v}_i)^2$, the fraction of the direction's norm lying in the rank-k principal subspace, for all k from 1 to d. The reference for these curves is a random unit direction, for which $\mathbb{E}[\rho_k] = k/d$.

Single-axis alignment. The fraction of features whose absolute cosine similarity with at least one of the leading k axes exceeds τ, reported jointly over $k \in \{1, \ldots, d\}$ and $\tau \in \{0.3, 0.5, 0.7, 0.9\}$ so that the figure does not depend on either choice.

Annotation yield per direction. The leading 50 principal axes were annotated with the identical top-20-gene enrichment protocol used for SAE features. Because a principal axis is signed whereas a TopK feature is one-sided, each axis was annotated in both polarities (genes with the most positive and most negative mean projection), and results are reported per polarity and for their union.

Counting distinct concepts across layers

Dictionary atoms from different layers occupy different vector spaces and cannot be pooled as directions. They can be compared in the shared gene space through their top-20 gene signatures. We built a similarity graph over all 82,525 Geneformer features, placing an edge between two features whose top-20 gene sets share at least 5 genes and have Jaccard similarity at least 0.14, and partitioned it with the Leiden algorithm [19] at resolutions $\gamma \in \{0.5, 1.0, 2.0\}$. The number of communities is the number of distinct gene-level programs. We additionally report, per layer, the number of atoms, the number of distinct programs, the mean absolute decoder coherence, and the Welch bound $\sqrt{(n-d)/(d(n-1))}$ for n unit vectors in $\mathbb{R}^d$, which is the smallest achievable maximum coherence and therefore the correct reference for how many near-orthogonal directions a d-dimensional space can host.

Co-activation graph and module detection

For each layer, pointwise mutual information (PMI [18]) was computed between all pairs of alive features across the layer's token positions; feature i is active at position j if it is among the top-k activations there, and $\mathrm{PMI}(i,j) = \log_2 P(i,j)/[P(i)P(j)]$. Edges were retained where two features co-fired at a minimum of 50 positions with PMI at least 2.0 bits. Community detection used the Leiden algorithm [19] at resolution $\gamma = 1.0$, the `leidenalg` default. Because a single resolution is a choice rather than a ground truth, we swept $\gamma \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$ at layers 0 and 11 and report the resulting partition sizes and adjusted Rand indices; module counts in the main text are the γ=1.0 cut of a hierarchical structure whose qualitative module identities are recognizable throughout the range.

Causal feature patching

Single-feature ablation was performed with PyTorch forward hooks. At the target layer's output the hidden state was encoded through the SAE and the contribution of one feature, its activation times its decoder atom, was subtracted from the residual stream; the forward pass then continued to the output. Because the intervention is subtractive rather than a substitution of the reconstruction, no reconstruction error is introduced. The readout is the logit of each position's own gene token. Disruption is the mean change in output logits at a set of gene positions.

Three properties of the design keep the evaluation independent of the annotation. First, features are drawn from three arms: the most richly annotated features, a uniformly random sample of annotated features, and a uniformly random sample of all alive features. Second, for each feature we partition gene positions into those whose gene is in both the feature's top-20 activating genes and its annotated term, those in the annotated term but *not* in the top-20 (the independent evaluation set), and all others. Third, for every feature we draw a random gene set matched in size to the independent evaluation set and stratified on gene expression rank (ten bins) and detection frequency across the patched cells (three bins), drawn from outside both the annotated term and the feature's top-20 genes, giving each feature its own null. Specificity is the ratio of mean absolute disruption on a gene set to that on all other positions, and the held-out ratio is reported against the matched-random ratio for the same feature.

Perturbation response mapping

For each perturbation target we sampled up to 200 CRISPRi cells and compared them against 600 non-targeting control cells of the same line, encoding every cell through the layer-11 SAE as described above. A feature responds if its per-cell mean activation differs between perturbed and control cells (Mann–Whitney U over cells, Benjamini–Hochberg across features, FDR < 0.05) and the shift exceeds half a control standard deviation, the latter measured on the position-level scale so that the cell-level and position-level analyses differ only in the inference. The cell is the unit of replication throughout: gene positions within a cell are not independent observations, and we report the intraclass correlation of feature activation within cells together with the number of features that a position-pooled test would call significant on exactly the same cells.

TF specificity is evaluated in the same framework as every other method (below): the union of the top-20 gene lists of responding features, ranked by effect size, forms the predicted target set, which is tested for enrichment of the TF's curated targets by hypergeometric test with Benjamini–Hochberg correction across the panel. A label-permutation null, in which perturbed and control labels are shuffled across the same cells, calibrates the procedure.

Recoverability benchmark

A recovery rate is interpretable only against the amount of information the measurements contain. We therefore evaluated several methods on one footing: each emits a ranked list of putative targets for every TF in the panel, the top 100 of which are tested for enrichment of that TF's curated targets by hypergeometric test with Benjamini–Hochberg correction across the panel.

Panel. TFs perturbed in the cell line with at least 50 cells and at least 5 curated targets present among the 6,546 measured genes. The second criterion is reported explicitly, because a TF whose curated targets are not measured cannot show target-specific behaviour under any method.

Universe. The hypergeometric universe is the set of measured genes. Using a nominal 20,000-gene universe is anticonservative here, since every method can only draw predictions from measured genes; we report both, and the random-gene-set control is calibrated only under the measured-gene universe.

Methods. (i) Differential expression of the knockdown itself (Mann–Whitney over cells, perturbed vs non-targeting, Benjamini–Hochberg across genes), which is perturbation-aware and constitutes the empirical ceiling attainable on these data. (ii) GENIE3 [30] and (iii) GRNBoost2 [31], in their published formulation: every target gene is regressed on the candidate regulators with a tree ensemble and regulators are ranked by variable importance (random forest with `max_features` $= \sqrt{p}$; and stochastic gradient boosting with learning rate 0.01, `max_features` $= 0.1$, subsample 0.9 and early stopping), fitted on the same control cells. (iv) Pearson and (v) Spearman co-expression with the TF on the same control cells. (vi) Cosine similarity between mean contextual gene embeddings taken from the Geneformer residual stream, a model-internal baseline that uses no SAE. (vii) Size-matched random gene sets. (viii) The SAE feature response described above.

Replication across cell lines

The perturbation analysis was repeated in RPE1, Jurkat and HepG2 cells from the same CRISPRi resource, which are independent screens in independent cell backgrounds. These lines were run at 100 cells per target against 400 non-targeting cells over 16 factors and 6 non-regulatory controls, against 200, 600, 20 and 10 respectively for the primary line. In every line, perturbed cells are compared against non-targeting cells of that same line. Each line was analysed with two dictionaries: the multi-line control atlas dictionary, and the dictionary additionally trained on Tabula Sapiens primary tissue, which asks whether widening the training distribution beyond immortalized lines changes what is recovered.

Independent external perturbation datasets

The measurement was additionally repeated on two perturbation datasets generated by other groups. ECCITE-seq of THP-1 monocytes [32] measures 18,649 genes; its perturbation labels are guide-level and were reduced to the targeted gene. Perturb-seq of K562 [33] measures 33,694 genes and is a CRISPR-activation screen, so it tests the same question under gene induction rather than repression; combinatorial perturbations were excluded. Counts were CP10K-normalized and log1p-transformed as for the primary resource, and genes were mapped to the model vocabulary by symbol, which covers 13,601 of 18,649 and 18,121 of 33,694 genes respectively.

Each dataset was analysed with the atlas dictionary at layer 11 under the pipeline described above: 100 cells per target against 400 non-targeting cells, cell-level Mann–Whitney with Benjamini–Hochberg across features, and the same hypergeometric enrichment of a 100-gene predicted target set. Panels comprise the perturbed transcription factors with at least 50 cells and at least five curated targets among the measured genes.

For the specificity control, ten perturbations of genes absent from every reference network were drawn at random from the same dataset and run through the identical pipeline. Their predicted sets were then tested against the *same* regulons as the transcription factors, giving one enrichment test per control-gene-by-regulon pair, with Benjamini–Hochberg applied within each arm. Because a control gene contributes many tests, the comparison between arms is made at the level of independent perturbations—how many control genes enrich for at least one regulon against how many transcription factors enrich for their own—by Fisher's exact test.

Run-to-run stability

The atlas SAE was retrained at layers 0 and 11 under five independent seeds with the training subsample held fixed, isolating initialization and batch-order stochasticity, plus one additional run per layer with a different 1M-position training subsample. Dictionaries were compared by decoder cosine similarity, reporting both the unconstrained best match for each atom and a greedy one-to-one assignment. Variance explained, dead-feature count, decoder coherence, annotation rate, and module count were recomputed per run.

Matched cross-model comparison

Because the two atlases were built from different inputs, architecture and input distribution are confounded in any direct comparison of their statistics. We therefore trained Geneformer SAEs on the same Tabula Sapiens cells used for the scGPT atlas, with identical hyperparameters, and compared the two models on matched data at matched relative depth (layer index divided by depth minus one). For the Geneformer arm we additionally separate two effects by reporting the variance explained when the control-cohort SAE is applied to Tabula Sapiens activations against that of the SAE trained on those activations directly.

Interactive web atlases

Both feature atlases were deployed as single-page web applications built with React 18, TypeScript, Vite 6, Tailwind CSS, and Plotly.js. All feature data were preprocessed into compact JSON files served statically via GitHub Pages.

Dimensionality reduction and visualization

We projected decoder weight vectors to two dimensions using UMAP [34] with cosine distance, obtaining structureless point clouds as expected from quasi-orthogonal directions. We therefore visualized the co-activation graph directly with a force-directed layout (Fruchterman–Reingold via NetworkX); the corresponding supplementary figures are force-directed layouts of that graph and are not UMAP or t-SNE projections. For independent validation, annotation-based feature vectors (TF-IDF weighted ontology terms) were projected with UMAP ($n_{\text{neighbors}} = 15$, $\text{min}_{\text{dist}} = 0.1$, cosine metric) and t-SNE (perplexity 20, cosine metric).

Computational environment

All experiments were run on a single Apple Silicon workstation (Apple M2 Pro, 10 cores, 32 GB unified memory). Acceleration used PyTorch's MPS backend, with CPU fallback for operations without an MPS implementation (Leiden community detection, ontology Fisher tests, tree-ensemble regression). Software stack: Python 3.11; PyTorch 2.x with MPS; NumPy, SciPy, scikit-learn, pandas, scanpy, anndata, leidenalg, python-igraph, networkx, umap-learn, statsmodels, h5py. Exact package versions are pinned in `requirements.txt` in the code repository. Wall-clock times for the longest steps are logged alongside the compute environment: Geneformer control-cohort activation extraction $\sim$8 h, SAE training $\sim$3 min per layer, per-layer PMI and Leiden $\sim$20–60 min, per-cell perturbation encoding $\sim$0.9 cells s^{-1}.

Data availability

All datasets analysed in this study are publicly available.

- **Genome-scale CRISPRi Perturb-seq** (K562, RPE1, Jurkat, HepG2) [12]: Figshare+, `https://doi.org/10.25452/figshare.plus.20029387.v1`.
- **ECCITE-seq of THP-1 monocytes** [32]: available through the scPerturb collection, `https://doi.org/10.5281/zenodo.13350497`.
- **Perturb-seq of K562** [33]: available through the scPerturb collection, `https://doi.org/10.5281/zenodo.13350497`.

- **Tabula Sapiens** [28]: `https://tabula-sapiens-portal.ds.czbiohub.org/` and CZ CELLxGENE.
- **Geneformer V2-316M** [1]: `https://huggingface.co/ctheodoris/Geneformer` (subfolder `Geneformer-V2-316M`).
- **scGPT whole-human** [2]: `https://github.com/bowang-lab/scGPT`.
- **TRRUST v2** [17]: `https://www.grnpedia.org/trrust/`.
- **DoRothEA** [29]: `https://saezlab.github.io/dorothea/`.
- **Gene Ontology** [13]: `http://geneontology.org/`.
- **KEGG** [14]: `https://www.kegg.jp/`. KEGG pathway gene sets are used under permission from Kanehisa Laboratories.
- **Reactome** [15]: `https://reactome.org/`.
- **STRING** [16]: `https://string-db.org/`.

The two feature atlases produced by this work are released as interactive web platforms: Geneformer (`https://biodyn-ai.github.io/geneformer-atlas/`) and scGPT (`https://biodyn-ai.github.io/scgpt-atlas/`).

Code availability

All analysis code, trained sparse autoencoder checkpoints, feature catalogs, and the result files from which every figure and table in this article is generated are archived at Zenodo (`https://doi.org/10.5281/zenodo.22068864`, doi:10.5281/zenodo.22068864), and are also available at `https://github.com/Biodyn-AI/bio-sae`. The archive includes the figure- and table-generation scripts, so each reported number can be traced to the result file that produced it, together with a reference-consistency check that verifies the correspondence between the main text and the Supplementary Information. Source for the interactive atlases is at `https://github.com/Biodyn-AI/geneformer-atlas` and `https://github.com/Biodyn-AI/scgpt-atlas`.

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Computations were performed on Apple Silicon hardware with MPS acceleration.

Author contributions

I.K. is the sole author. I.K. conceptualized the study, developed the methodology, wrote all software, performed all computational experiments, built the interactive feature atlases, analysed and interpreted the results, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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